smartoptics

Open Line Systems with ROADMs from the DCP-R Family

A solution brief from Smartoptics

Using ROADMs is since long the standard method to enable flexible add/drop of wavelengths in optical networks having ring and mesh topologies. With its open architecture and support for multiple traffic formats, the Smartoptics DCP-R Family allows you to implement virtually any type of ROADM-based network topology, disintegrating the once monolithic optical platforms into separate, open line systems, switches with embedded transceivers and transponders.

Smartoptics ROADM Portfolio

Being a pioneer in open line systems, it is a natural step for Smartoptics to also offer a portfolio of ROADMs using the same open and disaggregated architecture. If you want to leverage the current breakthroughs in DWDM transceiver technologies and associated cost reductions, it is a must to go for an open approach where the transceivers are embedded in standard switches and where the routing of the optical wavelengths is done by an open line system.

ROADM-based metro/regional and metro access networks can easily be built by the Smartoptics DCP-F family of versatile active optical units, by the dedicated DCP-R family ROADMs or by a combination of both families.

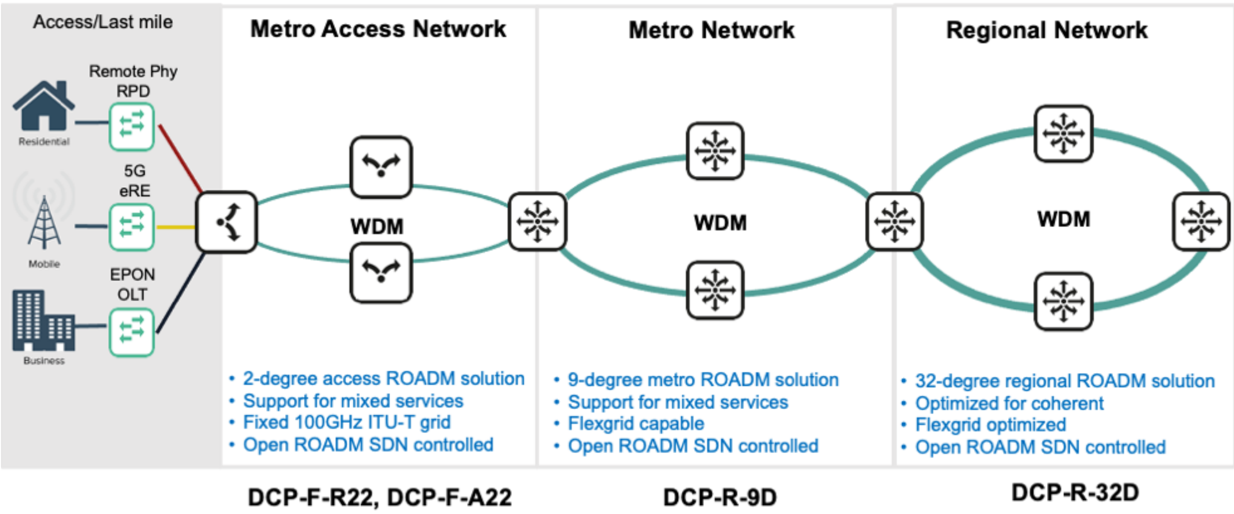

Figure 1. Smartoptics ROADM Portfolio

The DCP-F-R22 and DCP-F-A22 are two extremely versatile active optical units for configuration of any type of optical network at the lowest possible cost. Their main application as ROADMs is as 2-degree ROADMs in metro access ring networks and generally where low cost and seamless interworking with the passive parts of the network are a prerequisite for deployment.

The DCP-R-nD is a family of all-in-one ROADMs for 2- to 32-degrees, spanning applications from the metro access network up to large regional networks. The DCP-R family comprises compact, 1U per degree ROADMs having Flexgrid support and colorless, directionless and contentionless capabilities. The ROADMs, with integrated mux/demux for local add/drop, are true multipurpose units, designed for use with 100G PAM4 and 400ZR transceivers as well as with legacy 100G QPSK, 200G 16QAM, Ethernet and Fibre Channel traffic formats. The DCP-R ROADMs come in the same compact chassis as the well-known Smartoptics DCP-M family.

The DCP-R Family

The DCP-R family comprises the 9-degree ROADM DCP-R-9D and the 32-degree ROADM DCP-R-32D. The DCP-R-9D comes in two models, MS for Mixed Services, and CS for Coherent Services. The MS model being specially designed for combined use of the new PAM4 and 400ZR traffic formats with Ethernet, Fibre Channel, and standard coherent modulation schemes, and the CS model being optimized for coherent traffic formats.

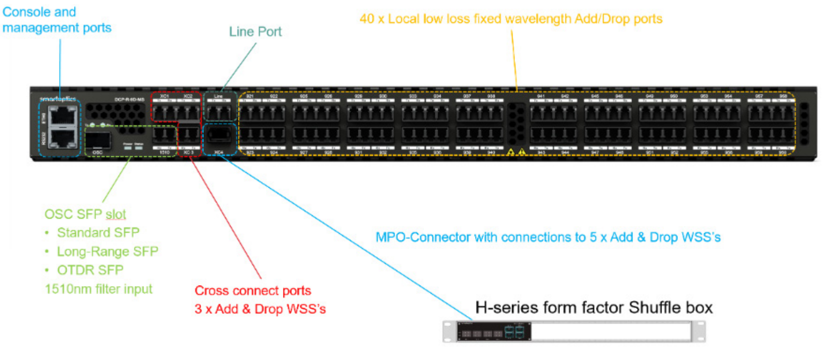

Figure 3. The DCP-R-9D chassis

All members of the DCP-R family support a high level of automation and openness. The ROADMs are typically controlled through the NetConf protocol compliant with the principles outlined by the OpenROADM MSA architecture and with TransportPCE as the SDN controller. The configuration of the ROADMs is further simplified by the integrated automatic fiber distance measurement and dispersion compensation setting. The DCP-R family supports a mixture of traffic formats with completely automatic detection of the modulation used and have Flexgrid, directionless, contentionless and colorless capabilities, all built on a scalable and cost effective datacom platform.

Traffic Formats and Modulation Support

A prominent characteristic of the DCP-R family is the great flexibility with which traffic formats can be mixed within the same ROADM. With the MS model, both direct detect and coherent formats can be supported simultaneously, alternatively the CS model, optimized for coherent formats, and providing more topology alternatives, can be chosen. This unique flexibility of the DCP-R family allows for efficient handling of legacy formats while offering an excellent upgrade path when new traffic formats based on new technologies are introduced in the network.

The achievable optical reach for each ROADM is traffic format dependent as well as DCP-R model dependent. Further information on the optical performance of the DCP-R family can be found in the data sheet for each model.

The DCP-R-9D ROADMs support the following traffic formats:

Traffic format	Supported by DCP-R-9D-MS	Supported by DCP-R-9D-CS
PAM4 (Colorz) modulated wavelengths (40G & 100G)	Yes	
400ZR OIF coherent modulated wavelengths (400G)	Yes	Yes
NRZ modulated wavelengths (1-10G Ethernet, 1-32G Fibre Channel, CPRI 1-10, SONET, SDH, OTN etc.)	Yes	Limited reach
Other coherent modulated wavelengths (100G, 200G & 400G)	Yes	Yes

ROADM Benefits

ROADMs (Reconfigurable Optical Add/Drop Multiplexers) are used in bus, ring and mesh shaped optical networks to enable flexible add/drop of individual wavelengths and for adding of new wavelengths without affecting the traffic on adjacent channels. The flexibility of ROADMs thus benefits the operator wanting to adapt to changing service requirements as well as overall network availability by simplifying protection switching and restoration of light paths. ROADM nodes also bring significant advantages from a network planning perspective. The free wavelength allocation with ROADMs simplifies traffic engineering and thereby reduces the effects of inaccurate traffic forecasting. In addition, ROADMs can perform automatic power balancing and simplify the structure of the physical network, thereby greatly simplifying fiber management.

ROADMs are typically designed around 1 x n Wavelength Selectable Switches (WSS) and multiplexers, as shown in the illustration. The number of optical fiber link directions to/from a ROADM is commonly referred to as the degree of the ROADM, e.g. the add/drop nodes of a ring network are often 2-degree ROADMs. Further flexibility can be given to the ROADM by making the mux/demux ports wavelength independent, i.e. making it possible to add/drop any wavelength at any port, creating a colorless ROADM. By adding further components the ROADM can allow for switching of any wavelength in either the “east” or “west” direction (directionless ROADM) and for the re-use of wavelengths that has been dropped in one direction in the continued light path (contentionless ROADM). Modern ROADMs are often capable of all the above, being Colorless, Directionless and Contentionless (CDC) at the same time.

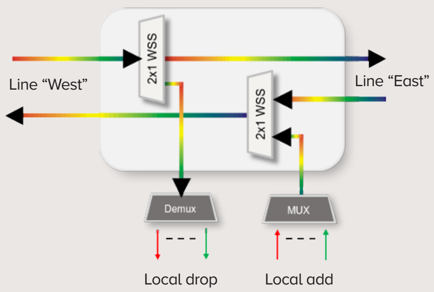

Figure 2. The main components of a 2-degree ROADM

Functional Design

DCP-R-9D-MS is designed especially for being capable of handling also lower output power transceivers such as ColorZ 100G PAM4 and 400ZR, typically used for high speed links at shorter distances. It constitutes an extremely compact, 1 U per degree ROADM with a built-in 40 channels mux/demux for local add/drop. Additional colorless add/drop ports may easily be added through an external Smartoptics H-series fiber shuffle box.

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The following illustration shows the block diagram of the DCP-R-9D-MS ROADM.

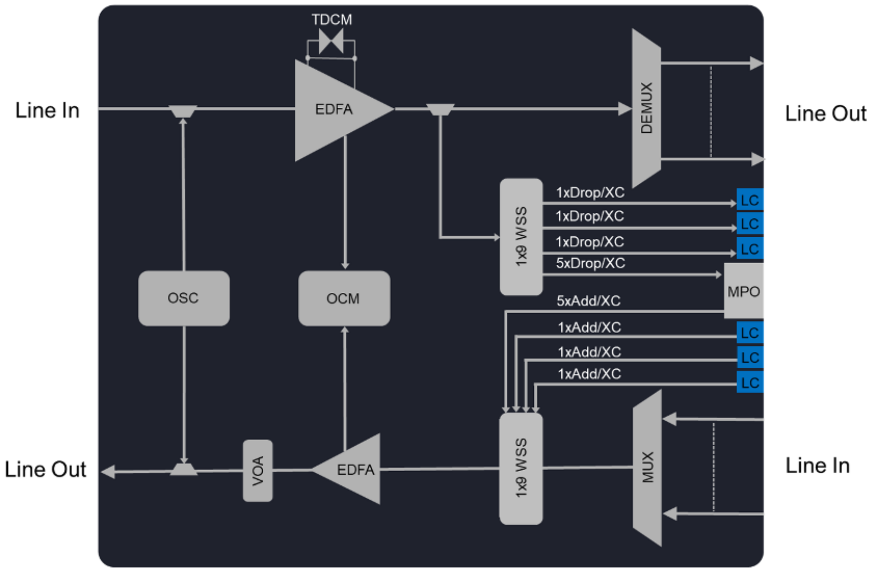
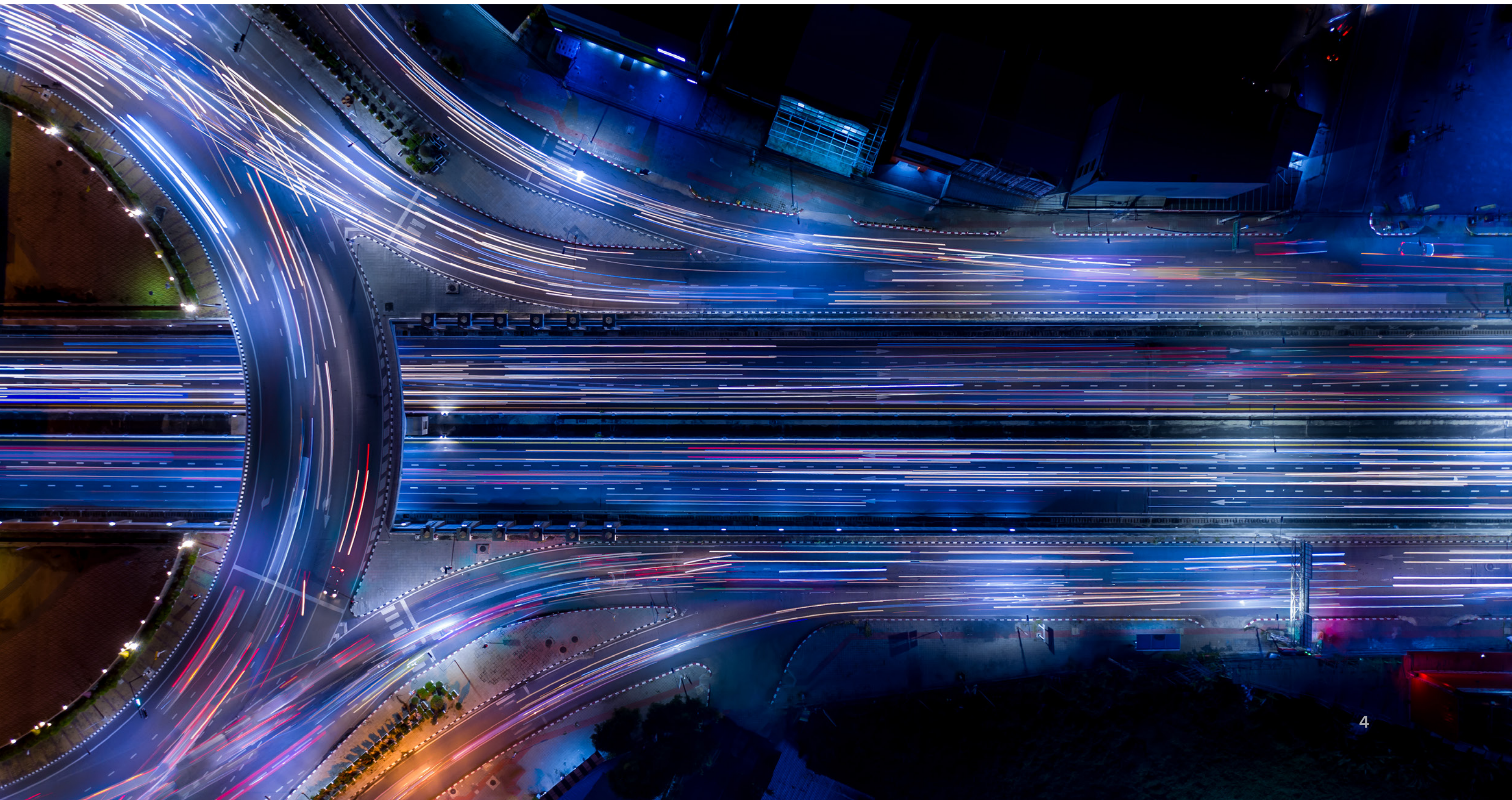


Figure 4. Block diagram of the DCP-R-9D-MS

The DCP-R-9D-CS ROADM has a similar design but without the dispersion control module (TDCM), which is not needed for the coherent modulation schemes.

Management of the DCP-R-9D, as well as for all the members of the DCP-R family, is either done by an SDN controller through the OpenROADM based API or manually by parameter settings from a command line interface (CLI).



2- and 4-degree ROADM Configurations

When the DCP-R-9D is used in a typical 2-degree ROADM configuration, two of the ports of the WSS are used for the optical connection to the next ROADM in the ring, as shown in the following illustration.

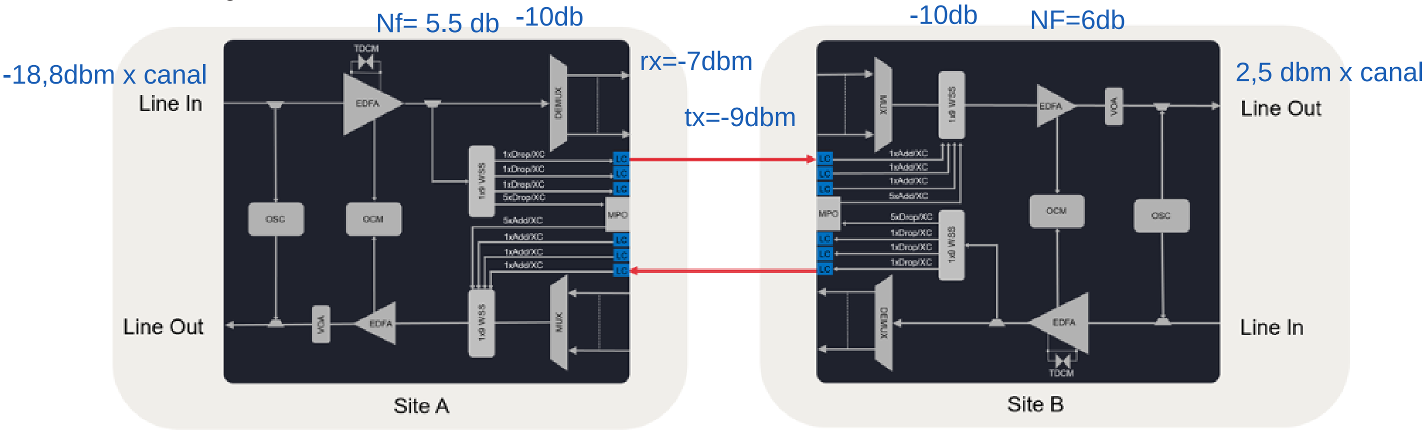


Figure 5. Typical 2-degree ROADM configuration with DCP-R-9D-MS

The remaining seven WSS ports of DCP-R-9D (two accessible via LC connectors and five via an MPO connector) can be used for:

- Direct connection of coherent wavelengths to the WSS
- Connection to a colorless filter
- Connection to a full Connectionless, Directionless, Colorless (CDC) filter (Amplification may be required depending on number of channels.)
- Flexgrid support to allow for future services with wider spectrum

The following illustration shows a complete ring where the DCP-R-9D-MS is used as a 2-degree ROADM. Due to the multiformat support and automatic modulation detection of the DCP-R-9D it is extremely easy for the operator to offer and manage multiple services with different traffic formats between individual add/drop sites.

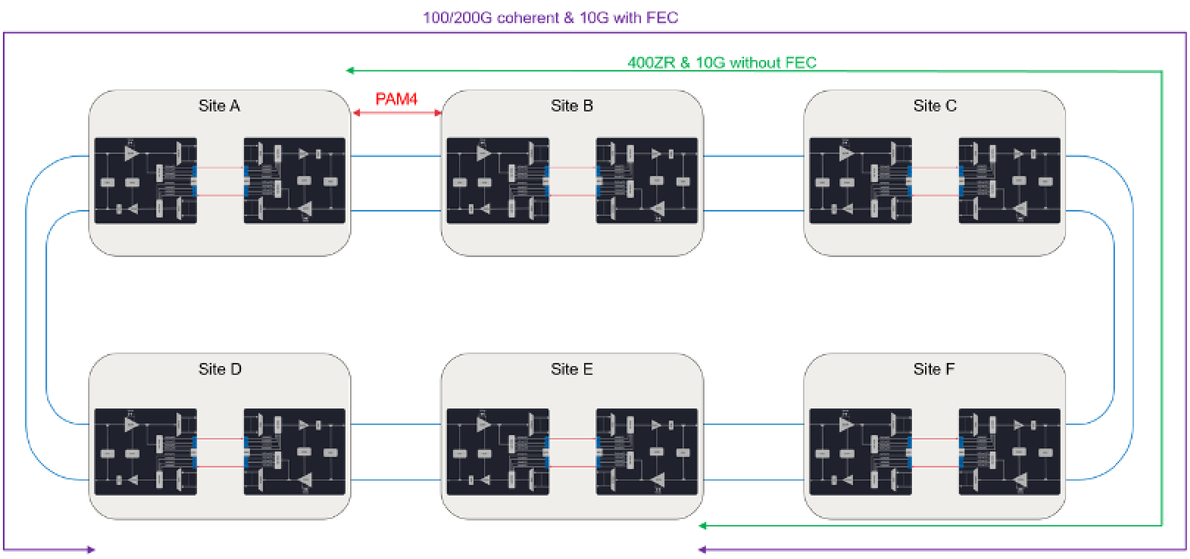


Figure 6. A complete ROADM ring with support for multiple traffic formats between individual sites

When creating a 4-degree ROADM site, four DCP-R-9D ROADMs are interconnected by use of a passive Smartoptics H-Series fiber shuffle box, as depicted in the next illustration. For each direction three additional WSS ports are available via LC ports on each ROADM front and two more via the shuffle box.

As for the 2-degree ROADM case, the free WSS ports may be used for direct connection of coherent wavelengths for colorless operation or for connections to a colorless or full CDC filter.

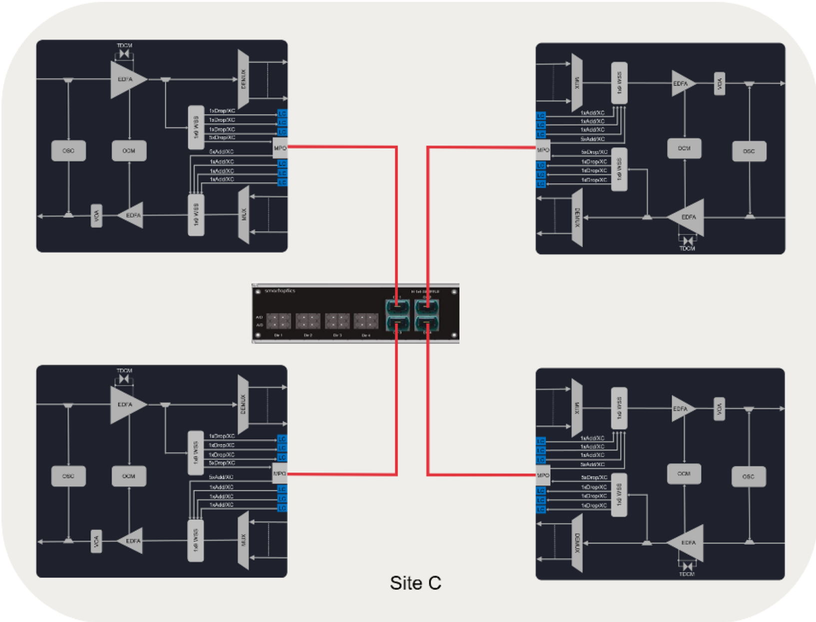


Figure 7. A 4-degree ROADM site with DCP-R-9D-MS

Access, Metro and Regional Networks

The DCP-R family ROADMs can easily be combined with the active optical units from the versatile DCP-F family, for example when deploying a combined metro and metro access optical network. As shown in the illustration below, DCP-R-9D ROADMs form the principal metro ring and the access rings are then connected to the metro ring by use of a 4-degree ROADM, also implemented by DCP-R-9D. Since all network elements use the same DCP platform from Smartoptics, management is greatly simplified while maintaining the open line systems architecture.

For additional information on the DCP-F family, see The DCP-F Family Solution Brief, available at https://www.smartoptics.com/

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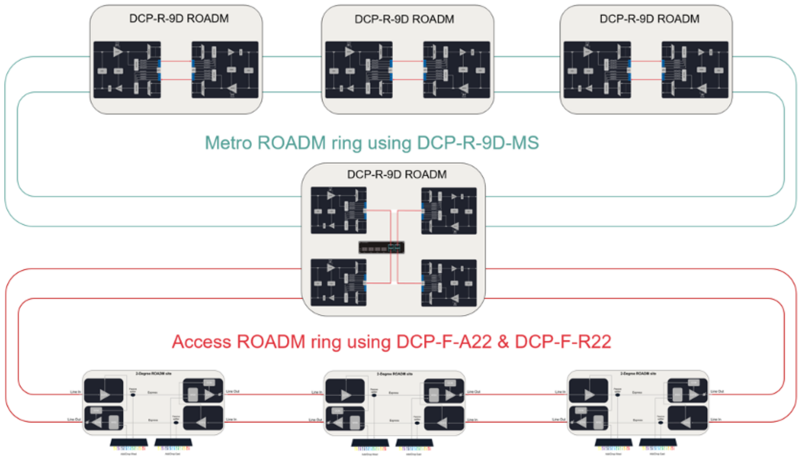

Figure 8. A metro and metro access network implemented by use of DCP-R-9D-MS, DCP-F-A22 and DCP-F-R22.

Another example of combined use of DCP-R and DCP-F family products is when longer distances need to be bridged and amplification of the optical signal is required. The dedicated DCP-F-A22 Optical Line Amplifier (OLA) with the necessary EDFA module can then be inserted as signal regenerator at suitable intermediate locations.

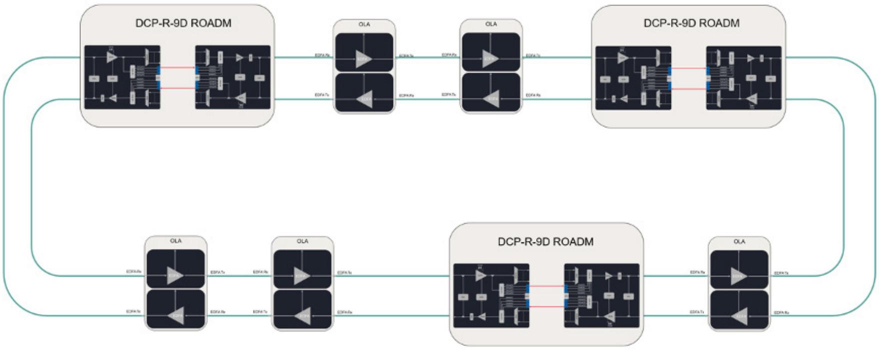

Figure 9. An example using DCP-F-A22 used for signal amplification and regeneration

Network and Wavelength Management

Looking at the total life cycle cost of a communications network, the recurring operational and management expenses dominate. Capable network management tools and network equipment designed for cost efficient maintenance are vital to make the communication network investment profitable and the business case attractive.

An open optical network requires a similarly open approach to network management. Where traditional optical transport system vendors have built proprietary software architectures for management of their own hardware, Smartoptics products provide open APIs that allow the customer to leverage open source solutions available today, develop their own applications on open SDN platforms or buy complete management solutions from other commercial players.

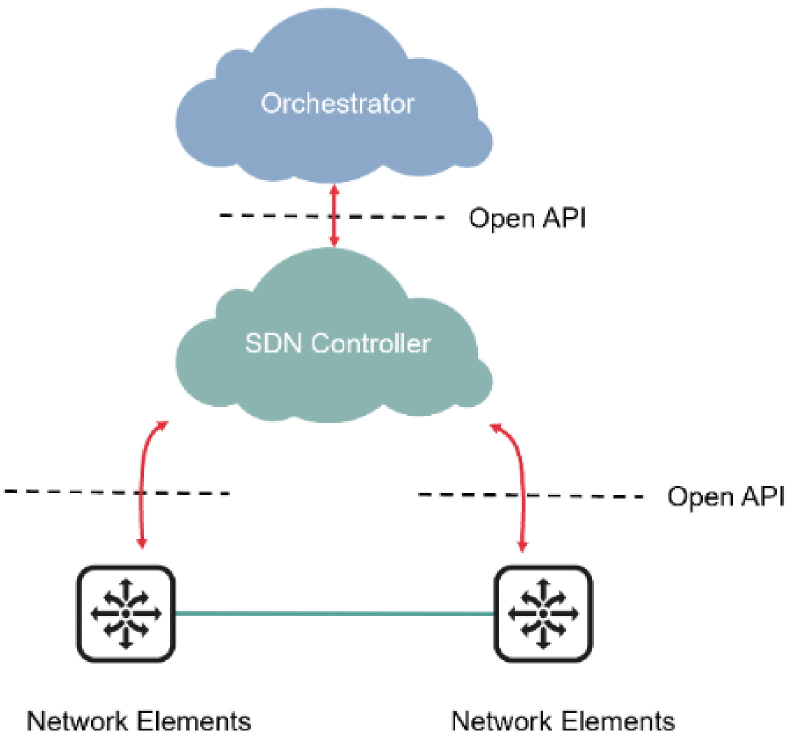

Figure 10. Smartoptics open network management architecture

An open optical network requires a similarly open approach to network management.

Our open API approach to network management brings several advantages. Rather than being dependent on the functionality and evolution of any vendor specific management platform, you may now leverage the more rapid advances of an open source effort such as the TransportPCE SDN controllers. Furthermore, an open management approach radically simplifies the integration of network elements from multiple vendors under the same SDN controller, thereby reducing the risk of any vendor lock-in. It also gives you a much higher degree of flexibility in designing the network management solution and adopting it to the optimal workflow for your organization. With open management APIs in the network elements you can go for anything from a sophisticated integration with your own SDN based applications to using a simple command line interface. Any development of customer specific software to manage a network is thus considerably less complicated in this modern and open environment than for a closed architecture, which is restricted by proprietary software functions tightly embedded in the network elements.

A Focus on Automation

Smartoptics traditional focus has been on the corporate data center interconnect (DCI) market where a high level of automation (plug&play) is greatly appreciated by our customers. Automation aspects are therefore particularly important to us and we are convinced that such operational benefits are of value to all our customers. We are hence designing for as much automation capabilities as possible within our products, both when used “stand alone” and in an SDN framework.

Management Architecture and Implementation

The DCP-M family, primarily used in point-to-point networks for Data Center Interconnect (DCI), is optimized for zero-touch, fully self-configuring/regulating deployment, while the DCP-R and DCP-F families, often deployed in operator networks, are primarily designed to be externally managed. The external management can either be via a command line interface (CLI) or by using Software Defined Networking (SDN) principles based on the Open ROADM MSA initiative. Depending on customer requirements management may be done directly from the customer´s already existing SDN controller/orchestrator via the Open ROADM API's of the network elements, by the Smartoptics Open SDN controller and the customer´s orchestrator or by the full Smartoptics SoSmart software suite. The availability of zero-touch, CLI and SDN-based management for each individual DCP product is release dependent.

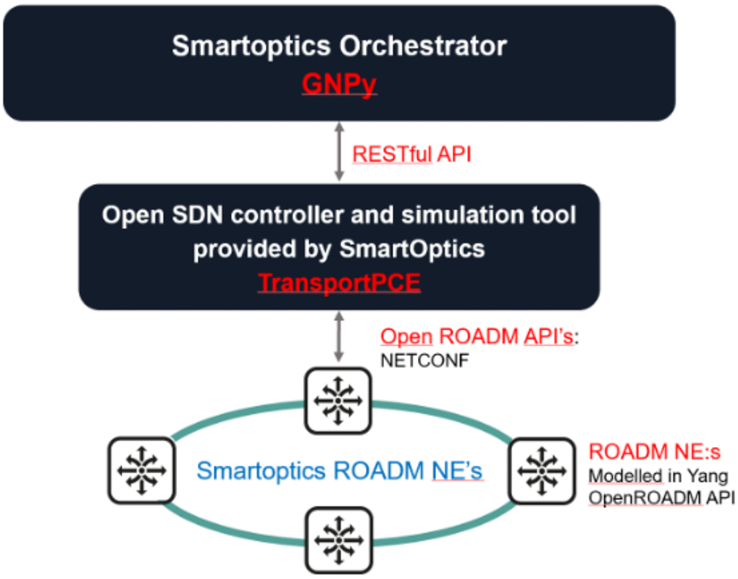

Figure 11. The SoSmart SDN software suite from Smartoptics

Smartoptics SoSmart:Classic and SoSmart:Cloud complete SDN software suites for open network management are released together with the introduction of the DCP-R family of ROADMs. These management suites, one server based and one cloud based, have a modern and hierarchical architecture that enables management flexibility, modularity, multiple integration possibilities and openness. The software suites include the following building blocks:

- Software embedded in the DCP network elements with data models based on Yang and supporting the Open ROADM API's, which are made accessible via the NetConf protocol
- An open source SDN controller based on TransportPCE
- The Smartoptics Open Optical Networking Orchestrator
- An open source offline optical simulation tool based on GNPy

Summary

The DCP-R family offers you a simple and straight forward way to leverage the benefits of open optical networking when implementing virtually any type of ROADM-based network topology. The family supports both new traffic formats using PAM4 and 400ZR modulation as well as legacy formats such as Ethernet, Fibre Channel, SONET, SDH and OTN including automatic fiber distance measurement and dispersion compensation settings. Modulation type is automatically detected, and the ROADM is controlled through an Open ROADM compliant API via NetConf, typically from an SDN controller. All DCP-R family members are designed for the highest level of quality and simplicity in use – a given choice for your network.

Product Source International Datacomm
330 Franklin Turnpike
Mahwah, NJ 07430
201.488.6000 Tel
www.psitec.com / sales@psitec.com

About Smartoptics

Smartoptics provides innovative optical networking solutions and devices for the new era of open networking. Our customer base includes thousands of enterprises, governments, cloud providers, Internet exchanges as well as cable and telecom operators. We have an open networking approach in everything we do which allows our customers to break unwanted vendor lock-in, remain flexible and minimize costs. Our solutions are used in metro and regional network applications that increasingly rely on data center services and specifications. Smartoptics is a Scandinavian company founded in 2006. We partner with leading technology and network solution providers such as Brocade, Cisco, HPE and Dell EMC and have a global reach through more than 100 business partners.

For additional information about Smartoptics, please visit https://www.smartoptics.com/

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Ryensvingen 7
NO-0680 Oslo
Norway
+47 21 41 74 00

info@smartopics.com
smartoptics.com

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